EET424 – ENERGY MANAGEMENT

EIGHTH SEMESTER EEE

Question & Answers Bank

MODULE 2

Energy Efficiency in Electricity Utilization:

Electricity transmission and distribution system, cascade efficiency.

Lighting: Modern energy efficient light sources, life and efficacy comparison with older light sources, energy conservation in lighting, use of sensors and lighting automation.

Motors: Development of energy efficient motors and the present status, techniques for improving energy efficiency, necessity for load matching and selection of motors for constant and variable loads.

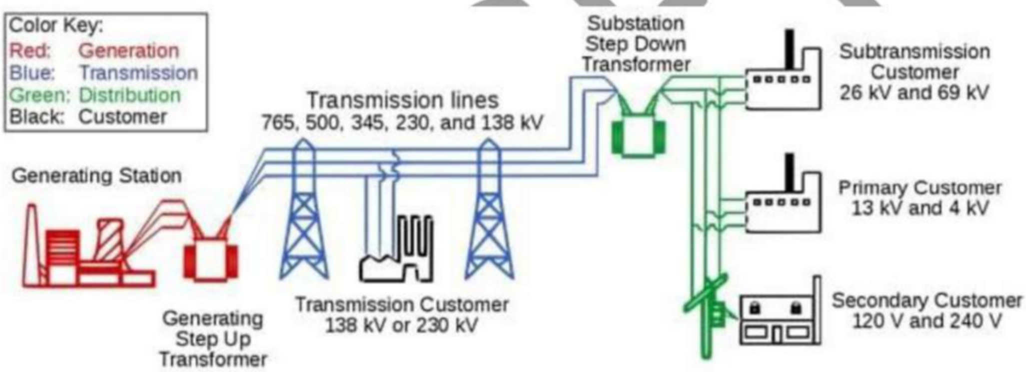
Transformers: Present maximum efficiency standards for power and distribution transformers, design measures for increasing efficiency in electrical system components.

Module 2

1. Explain about Utility Power Transmission and Distribution Systems. (10)

Electrical power used in residential, commercial, and industrial buildings is typically generated by a utility at a central point and transmitted and distributed to where it is required through the utility power transmission and distribution system.

A utility power transmission and distribution system controls, protects, transforms, and regulates electrical power so it can be safely delivered to the user. The utility power transmission and distribution system begins at the point of power production and normally ends at a building metered service entrance point, which is where the building distribution system begins. A utility power transmission and distribution system consists of transmission substations (step-up transformers), transmission lines, distribution substations (step-down transformers), and distribution lines.



Transmission Substations

A transmission substation is an outdoor facility located along with a utility system that is used to change voltage levels, provide a central place for system switching, monitoring, protection, and redistribute power. Transmission substations normally operate at high voltage (HV), 69 kV to 345 kV, and extra-high voltage (EHV), the voltage over 345 kV. Transmission substations are also used to make changes in the size and number of lines sent out from the station.

Transformers

The transformer is an electrical device that changes the voltage from one level to another by using electromagnetism. In electrical distribution systems, transformers are used to safely and efficiently increase or decrease voltage. Although a transformer can be used to increase or decrease voltage, transformers cannot be used to increase or decrease the amount of power available. Except for some minor power loss caused primarily by heat loss, the amount of power entering a transformer is the same amount of power leaving the transformer. Transformers are rated in kVA, which specifies their power output capability.

The main advantage of increasing voltage and reducing current is that power may be transmitted through small gauge conductors, which reduces the cost of power lines. For this reason, the generated voltages are stepped up to high levels for distribution across large distances and then stepped down to meet user requirements. Though both current and voltage can be stepped down or up, when it comes to transformers, the terms "step up" and "step down" always refer to voltage.

Transmission Lines

A transmission line is an aerial conductor that carries large amounts of electrical power at high voltages over long distances. To be safe, transmission lines must be positioned far enough apart. The transmission voltage level is determined by the required transmission distance as well as the amount of power carried. A larger transmission voltage is chosen when dealing with longer distances or larger transmitted power levels.

There is a wide variety of transmission line voltages, ranging from a few kilovolts to hundreds of kilovolts. Transmission-line voltage is stepped up to allow large amounts of power to be transmitted using smaller conductors. Since conductor sizes are based on the amount of current they can safely carry without overheating, low current levels can be carried over small size conductors.

In addition, increasing the transmitted voltage lowers the power losses between the utility generator and the final delivery point. Doubling the transmitted voltage can reduce the power loss by up to 75%. Because transmitting power at high voltages reduces the required size and weight of the conductors, the poles and towers that support the conductors can be smaller and spaced farther apart. Therefore, greater transmitted voltages allow for smaller conductor sizes, higher power transmission, and lower construction and material costs.

Distribution Substations

A distribution substation is fundamentally an outdoor facility that is located near the point of electrical service use and is used to adjust voltage levels, provide a central place for system monitoring, switching, and protection, and redistribute power. Distribution substations take high transmitted voltages and reduce the voltage for further distribution. Transmission substations operate at higher voltages, whereas distribution substations operate at lower voltages. The output voltages of distribution substations typically range from 12 kV to 13.8 kV.

Distribution substations provide a location along the distribution system near the end-user to easily test the system, adjust voltage output, add new lines, disconnect lines, and redirect power during distribution system problems such as power outages caused by lightning strikes. See Figure 5. Distribution substations take the incoming power and, after changing the voltage level, produce multiple outputs with different voltages on each line.

Distribution Lines

Distribution lines are used to carry electrical power from a distribution substation to the building service entrance. Distribution lines connect parts of the system together and are often run in multiple lines so that electrical power can be switched to meet changing power requirements and switched between different utilities. The term "grid" is used to describe the network of interconnected transmission and distribution lines.

2. Define cascade efficiency of an electrical system. How it can be calculated? (7)

The primary function of transmission and distribution equipment is to transfer power economically and reliably from one location to another.

Conductors in the form of wires and cables strung on towers and poles carry the high voltage A.C. current.

A large number of copper or aluminium conductors are used to form the transmission path.

The resistance of the long-distance transmission conductors is to be minimized. Energy loss in transmission line is wasted in the form of I2R losses.

Capacitors are used to correct power factor by causing the current to lead the voltage. When A.C. currents are kept with voltage, operating efficiency of the system is maintained at a high level.

Circuit interrupting devices are switches, relays, circuit breakers and fuses. Each of these devices is designed to carry and interrupt certain levels of current.

Making and breaking the current carrying conductors in the transmission path with a minimum of arcing is one of the most important characteristics of this device.

Relays sense abnormal voltages, currents and frequency and operate to protect the system.

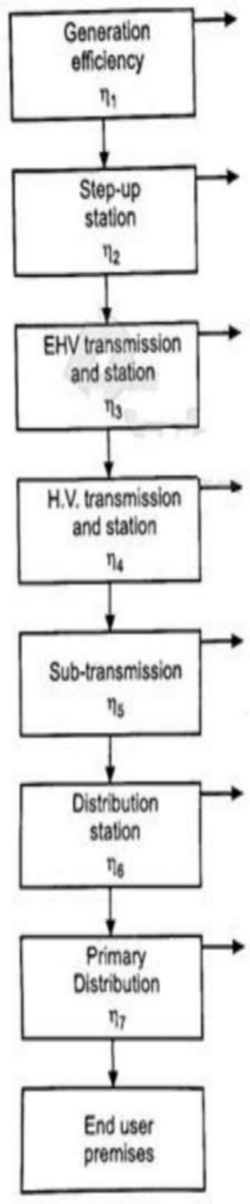
Transformers are placed at strategic locations throughout the system to minimize power losses in the transmission and distribution system.

They are used to change the voltage level from low to high or from high to low.

The power source to end user energy efficiency link is a key factor, which influences the energy input at the source of supply.

If we consider, the electricity flow from generation to the user in terms of cascade energy efficiency.

A typical cascade efficiency profile from generation to 11-33 kV user industry will be as follows.



Efficiency ranges 28 - 35% with respect to size of thermal plant, age of plant and capacity utilization

Step-up to 400/800 kV to enable EHV transmission.
Maximum losses 0.5% or efficiency 99.5%

EHV transmission and sub-station at 400 kV/800 kV.
Maximum losses 1.0% or efficiency 99%

HV transmission and sub-station for 220/400 kV.
Maximum losses 2.5% or efficiency 97.5%

Sub-transmission at 66/132 kV
Maximum losses 4% or efficiency 96%
Step-down to a level of 11/33 kV
Maximum losses 0.5% or efficiency 99.5%

Distribution is final link to end user at 11/33 kV.
Maximum losses 5% or efficiency 95%
Cascade efficiency from generation to end user
$= \eta_1 \times \eta_2 \times \eta_3 \times \eta_4 \times \eta_5 \times \eta_6 \times \eta_7$

The cascade efficiency in the Transmission and Distribution system from output of power plant to end user is calculated as follows:

Cascade Efficiency = 0.995 x 0.99 x 0.975 x 0.96 x 0.995 x 0.95 = 0.87 = 87%

3. Explain the Energy management opportunities in lighting system. (10)

4. Explain any five energy management opportunities in lighting system (5)

Electric lighting is a major energy consumer. Enormous energy savings are possible using energy efficient equipment, effective controls, and careful design. Using less electric lighting reduces heat gain, thus saving air-conditioning energy and improving thermal comfort. Electric lighting design also strongly affects visual performance and visual comfort by aiming to maintain adequate and appropriate illumination while controlling reflection and glare.

Lighting is not just a high priority when considering hotel design; it is also a high return, low-risk investment. By installing new lighting technologies, hotels can reduce the amount of electricity consumed and energy costs associatedwith lighting. There are several types of energy efficient lighting and affordable lighting technology.

The following are a few examples of energy-saving opportunities with efficient lighting.

1. Installation of compact fluorescent lamps (CFLs) in place of incandescent lamps.
2. Installation of energy-efficient fluorescent lamps in place of "conventional" fluorescent lamps.
3. Installation of occupancy/motion sensors to turn lights on and off where appropriate.
4. Use an automated device, such as a key tag system, to regulate the electric power in a room.
5. Offer nightlights to prevent the bathroom lights from being left on all night.
6. Replace all exit signs with light emitting diode (led) exit signs.
7. Use high efficiency (hid) exterior lighting
8. Add lighting controls such as photo sensors or time clocks

1. **Installation of compact fluorescent lamps (CFLs) in place of incandescent lamps**

Compact Fluorescent Lamps use a different, more advanced technology than incandescent light bulbs and come in a range of styles and sizes based on brand and purpose. They can replace regular, incandescent bulbs in almost any light fixture including globe lamps for the bathroom vanity, lamps for recessed lighting, dimming, and 3-way functionality lights. CFLs use about 2/3 less energy than standard incandescent bulbs, give the same amount of light, and can last 6 to 10 times longer. CFL prices range

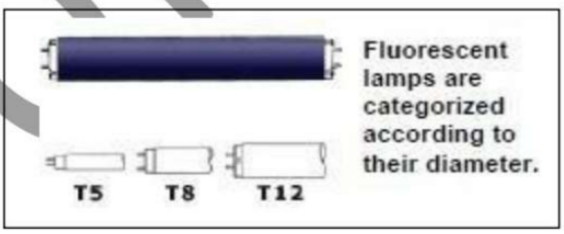
from $4 to $15 depending on the bulb, but you save about $25 to $30 per bulb on energy during the lifetime of the bulb.

When looking to purchase CFLs in place of incandescent bulbs, compare the light output, or Lumens, and not the watts. Watts refers to the amount of energy used, not the amount of light. In other words, if the incandescent bulb you wish to replace is 60 Watts, this is equal to 800 Lumens

To get the same amount of light in a CFL, you should look to find a CFL that provides 800 Lumens or more (equal to about a 13 watt fluorescent bulb).

Minimum light output (lumens)	Electrical power consumption (watts)		
	Incandescent	Compact fluorescent	LED
450	40	9–11	6–8
800	60	13–15	9–12
1,100	75	18–20	13–16
1,600	100	23–28	15–22
2,400	150	30–52	24–28
3,100	200	49–75	30
4,000	300	75–100	38

2. **Installation of energy-efficient fluorescent lamps in place of "Conventional" fluorescent lamps.**



Many lodging facilities may already use fluorescent lighting in their high traffic areas such as the lobby or office area. However, not all fluorescent lamps are energy efficient and cost effective. There are several types of fluorescent lamps that vary depending on the duration of their lamp life, energy efficiency, regulated power, and the quality of color it transmits.

There are a few styles worth noting; these models are simply labeled as "T-12", "T- 8", or "T-5". The names come from the size of their diameter per eighth inch. For example, a T-12 lamp is 12/8 inch in diameter (or 1 1/2 inch); a T-8 lamp is 8/8 inch in diameter (or 1 inch); a T-5 lamp is 5/8 inch in diameter. This is a simple way to identify the type of fluorescent lamps your facility is using.

The recommended style of fluorescent lighting is a T-8. T-8 lights are the most cost-effective. They usually cost about $0.99 a bulb and are 30% to 40% more efficient than standard T-12 fluorescent lamps, which have poor color rendition and cause eye strain. T-8 lamps provide more illumination, better color, and don't flicker (often exhibited by standard fluorescent fixtures). T-5 lamps are the most energy efficient and also tend to transmit the best color; however, they usually cost about $5.00 per bulb.

3. **Installation of occupancy/motion sensors to turn lights on and off where appropriate.**

Lighting can be controlled by occupancy sensors to allow operation whenever someone is within the area being scanned. When motion can no longer be detected, the lights shut off. Passive infrared sensors react to changes in motion. The controller must have an unobstructed view of the building area being scanned. Doors, partitions, stairways, etc. will block motion detection and reduce its effectiveness. The best applications for passive infrared occupancy sensors are open spaces with a clear view of the area being scanned.

Ultrasonic sensors transmit sound above the range of human hearing and monitor the time it takes for the sound waves to return. A break in the pattern caused by any motion in the area triggers the control.

Ultrasonic sensors can see around obstructions and are best for areas with cabinets and shelving, restrooms, and open areas requiring 360-degree coverage. Some occupancy sensors utilize both passive infrared and ultrasonic technology, but are usually more expensive. They can be used to control one lamp, one fixture or many fixtures. It can work in 3 modes: Time out, sensor mode and motion sensitivity settings The table below provides typical savings achievable for specific building areas, by the implementation of motion sensors.

Application	Potential Energy Cost Savings
Offices (private)	25-50%
Offices (open areas)	20-25%
Restrooms	30-75%
Corridors	30-40%
Storage areas	45-65%
Meeting rooms	45-65%
Conference rooms	45-65%
Warehouses	50-75%

4. Use an automated device, such as a key tag system, to regulate the electric power in a room.

The key tag system uses a master switch at the entrance of each guest room, requiring the use of a room key-card to activate them. Using this technique, only occupied rooms consume energy because most electrical appliances are switched off when the keycard is removed (when the guest leaves the room). The key tag system uses a master switch at the entrance of each guest room, requiring the use of a room key-card to activate them. Using this technique, only occupied rooms consume energy because most electrical appliances are switched off when the keycard is removed (when

the guest leaves the room). Along with lighting, the heating, air conditioning, radio and television may also be connected to the master switch. This innovation has a potential savings of about $105.00 per room per year.

5. Offer nightlights to prevent the bathroom lights from being left on all night

Many guests opt to have a light on while they sleep. By turning the bathroom light on and leaving the bathroom door cracked open, guests are able to find their way through an unknown room in the middle of the night. Those who are accompanied by children may often do the same to comfort their child. By offering a nightlight, the energy used to power a bathroom light during the nighttime can be avoided and guests will still be able to feel comfortable in unfamiliar territory.

One particular model uses six Light Emitting Diodes (LEDs) in the panel of a light switch to provide light for guests. LEDs are just tiny light bulbs that fit easily into an electrical circuit. They are different from ordinary incandescent bulbs because they don't burn out or get really hot. They are often used in digital clocks or remote controls.

6. Replace all exit signs with light emitting diode (LED) exit signs.

The development of light emitting diodes (LEDs) has allowed the replacement of exit sign lighting with a more energy efficient alternative. Multiple LEDs, properly configured, produce equivalent lighting and consume 95% less electricity than incandescent bulbs and compact fluorescent lamps is 75% less energy-efficient than LED.

A major benefit is the 20-year life cycle rating of LEDs; they virtually eliminate maintenance. Of the three different styles of exit signs, incandescent signs are the least expensive, but are inefficient and use energy releasing heat instead of light. Fluorescent signs are also inexpensive and have an expected life of about 10,000 hours.

LED exit signs are the most expensive, but are also the most efficient exit signs available. Their payback time is usually about four years. The table on the following page offers an easy comparison of the three models of exit signs.

	Incandescent	Fluorescent	LED
Input Power (watts)	40	11	2
Yearly energy (kWh)	350	96	18
Lamp life (years)	0.25-0.5	1-2	10+
Estimated energy cost/ year ($0.06/kWh)	$21.00	$5.75	$1.10

7. Use high efficiency (hid) exterior lighting

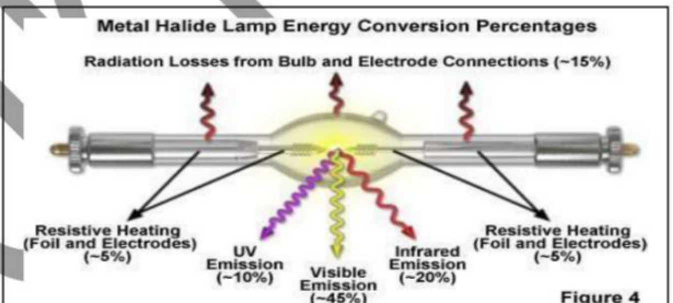


Figure 4

High intensity discharge (HID) lighting is much more efficient and preferable to incandescent, quartz-halogen and most fluorescent light fixtures. HID types (from least to most efficient) include mercury vapor, metal halide and high-pressure sodium. Mercury vapor is seldom used anymore.

Both metal halide and high-pressure sodium are excellent outdoor lighting systems. High pressure sodium has a pink-orange glow and is used when good color rendition isn't critical.

Metal halide, though less efficient, provides clean white light and good color rendition. HID lighting is mostly utilized in floodlight, wall pack, canopy and area fixtures outdoors. The best type for any application depends on the area being lit and mounting options.

8. Add lighting controls such as photo sensors or time clocks

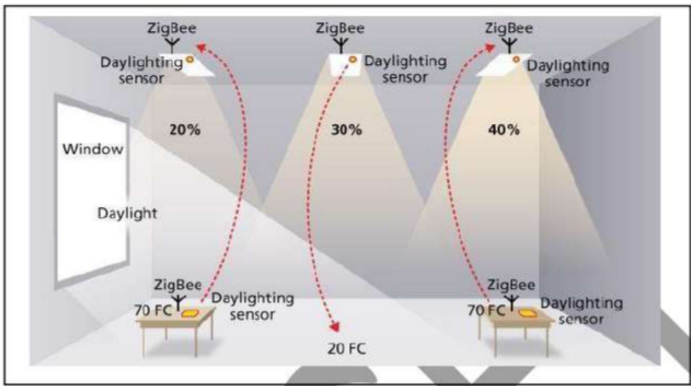


Photo sensor controls monitor daylight conditions and allow fixtures to operate only when needed. Photo sensors detect the quantity of light and send a signal to a main controller to adjust the lighting. Photo sensors are commonly used with outdoor lighting to automatically turn lights on at dusk and off at dawn, a very cost-effective control device. This helps to lower energy costs by ensuring that unnecessary lighting is not left on during daytime hours.

Photo sensors can be used indoors, as well. Building areas with lots of windows may not require lights to be on all of the time. Photocells can be used to ensure fixtures operate only when the natural light is inadequate by either controlling one light fixture, or a group of lights.

The table below demonstrates the cost savings from day light controls.

Potential Annual Lighting Energy Cost Savings with Day lighting Controls

Day lighting Strategy	Control Type	Potential Annual Energy Savings
Window sidelighting	On/off	32%
	Stepped	44%
	Continuous dimming	56%
Skylighting	On/off	52%
	Stepped	57%
	Continuous dimming	62%

Source: Lawrence Berkeley Laboratory

Time controls save energy by reducing lighting time of use through pre-programmed scheduling. Time clock equipment ranges from simple devices designed to control a single electrical load to sophisticated systems that control several lighting zones. They are one of the simplest, least expensive, and most efficient energy management devices available. Time controls could include:

Simple time switches: automatically turn lights, fans or other electronic devices off after a pre-set time.

Multi-channel time controls: have the ability to control from 4 to 16 duties.

Special-purpose time controls: include cycle timers for repetitive short duration cycling of equipment or outdoor lighting time controls that combine time clock and photo sensor technologies.

5. **Explain the Energy management opportunities in Motors. (10)**
6. **What are the factors which affect Energy Efficiency of motors and explain the steps to minimize Motor Losses in operation (5)**

Motors convert electrical energy into mechanical energy by the interaction between the magnetic fields set up in the stator and rotor windings. Industrial electric motors can be broadly classified as induction motors, direct current motors or synchronous motors. All motor types have the same operating components: stator (stationary windings), rotor (rotating windings), bearings, and frame (enclosure).

Two important attributes relating to efficiency of electricity use by A.C. Induction motors are efficiency (η), defined as the ratio of the mechanical energy delivered at the rotating shaft to the electrical energy input at its terminals, and power factor (PF). Motors, like other inductive loads, are characterized by power factors less than one. As a result, the total current draw needed to deliver the same real power is higher than for a load characterized by a higher PF. An important effect of operating with a PF less than one is that resistance losses in wiring upstream of the motor will be higher, since these are proportional to the square of the current. Thus, both a high value for η and a PF close to unity are desired for efficient overall operation in a plant.

Squirrel cage motors are normally more efficient than slip-ring motors, and higher-speed motors are normally more efficient than lower-speed motors. Efficiency is also a function of motor temperature. Totally-enclosed, fan-cooled (TEFC) motors are more efficient than screen protected, drip-proof (SPDP) motors. Also, as with most equipment, motor efficiency increases with the rated capacity. The efficiency of a motor is determined by intrinsic losses that can be reduced only by changes in motor design. Intrinsic losses are of two types: fixed losses - independent of motor load, and variable losses - dependent on load.

Energy-Efficient Motors

Energy-efficient motors (EEM) are the ones in which, design improvements are incorporated specifically to increase operating efficiency over motors of standard design (see Figure 2.3). Design improvements focus on reducing intrinsic motor losses. Improvements include the use of lower-loss silicon steel, a longer core (to increase active material), thicker wires (to reduce resistance), thinner laminations, smaller air gap between stator and rotor, copper instead of aluminum bars in the rotor, superior bearings and a smaller fan, etc.

Energy-efficient motors now available in India operate with efficiencies that are typically 3 to 4 percentage points higher than standard motors. In keeping with the stipulations of the BIS, energy-efficient motors are designed to operate without loss in efficiency at loads between 75 % and 100 % of rated capacity. This may result in major benefits in varying load applications. The power factor is about the same or may be higher than for standard motors. Furthermore, energy efficient motors have lower operating temperatures and noise levels, greater ability to accelerate higher-inertia loads, and are less affected by supply voltage fluctuations.

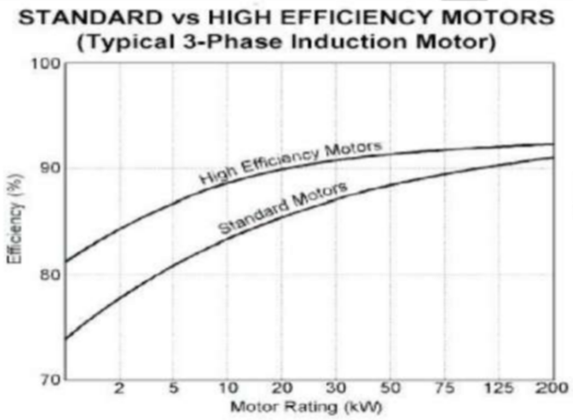


Figure 2.3 Standard vs High Efficiency Motors

Energy management opportunities in Motors

Stator and Rotor I^2R Losses

These losses are major losses and typically account for 55% to 60% of the total losses. I^2R losses are heating losses resulting from current passing through stator and rotor conductors. I^2R losses are the function of a conductor resistance, the square of current. Resistance of conductor is a function of conductor material, length and cross-sectional area. The suitable selection of copper conductor size will reduce the resistance. Reducing the motor current is most readily accomplished by decreasing the magnetizing component of current. This involves lowering the operating flux density and possible shortening of air gap. Rotor I^2R losses are a function of the rotor conductors (usually aluminum) and the rotor slip. Utilization of copper conductors will reduce the winding resistance. Motor operation closer to synchronous speed will also reduce rotor I^2R losses.

Core Losses

Core losses are those found in the stator-rotor magnetic steel and are due to hysteresis effect and eddy current effect during 50 Hz magnetization of the core material. These losses are independent of load and account for 20 – 25 % of the total losses. The hysteresis losses which are a function of flux density, are be reduced by utilizing low loss grade of silicon steel laminations. The reduction of flux density is achieved by suitable increase in the core length of stator and rotor. Eddy current losses are generated by circulating current within the core steel laminations. These are reduced by using thinner laminations.

Friction and Windage Losses

Friction and windage losses results from bearing friction, windage and circulating air through the motor and account for 8 – 12 % of total losses. These losses are independent of load. The reduction in heat generated by stator and rotor losses permit the use of smaller fan. The windage losses also reduce with the diameter of fan leading to reduction in windage losses.

Stray Load-Losses

These losses vary according to square of the load current and are caused by leakage flux induced by load currents in the laminations and account for 4 to 5 % of total losses. These losses are reduced by careful selection of slot numbers, tooth/slot geometry and air gap. Energy efficient motors cover a wide range of ratings and the full load efficiencies are higher by 3 to 7 %. The mounting dimensions are also maintained as per IS1231 to enable easy replacement.

As a result of the modifications to improve performance, the costs of energy-efficient motors are higher than those of standard motors. The higher cost will often be paid back rapidly in saved operating costs, particularly in new applications or end-of-life motor replacements. In cases where existing motors have not reached the end of their useful life, the economics will be less clearly positive. Also, most energy-efficient motors produced today are designed only for continuous duty cycle operation.

Given the tendency of over sizing on the one hand and ground realities like ; voltage, frequency variations, efficacy of rewinding in case of a burnout, on the other hand, benefits of EEM's can be achieved only by careful selection, implementation, operation and maintenance efforts of energy managers. A summary of energy efficiency improvements in EEMs is given in the Table 2.2

TABLE 2.2 ENERGY EFFICIENT MOTORS

Power Loss Area	Efficiency Improvement
1. Iron	Use of thinner gauge, lower loss core steel reduces eddy current losses. Longer core adds more steel to the design, which reduces losses due to lower operating flux densities.
2. Stator I^2R	Use of more copper and larger conductors increases cross sectional area of stator windings. This lowers resistance (R) of the windings and reduces losses due to current flow (I).
3. Rotor I^2R	Use of larger rotor conductor bars increases size of cross section, lowering conductor resistance (R) and losses due to current flow (I).
4. Friction & Windage	Use of low loss fan design reduces losses due to air movement.
5. Stray Load Loss	Use of optimized design and strict quality control procedures minimizes stray load losses.

Power Factor Correction

As noted earlier, induction motors are characterized by power factors less than unity, leading to lower overall efficiency (and higher overall operating cost) associated with a plant's electrical system. Capacitors connected in parallel (shunted) with the motor are typically used to improve the power factor. The impacts of PF correction include reduced kVA demand (and hence reduced utility demand charges), reduced I2R losses in cables upstream of the capacitor (and hence reduced energy charges), reduced voltage drop in the cables (leading to improved voltage regulation), and an increase in the overall efficiency of the plant electrical system.

It should be noted that PF capacitor improves power factor from the point of installation back to the generating side. It means that, if a PF capacitor is installed at the starter terminals of the motor, it won't improve the operating PF of the motor, but the PF from starter terminals to the power generating side will improve, i.e., the benefits of PF would be only on upstream side.

Maintenance

Inadequate maintenance of motors can significantly increase losses and lead to unreliable operation. For example, improper lubrication can cause increased friction in both the motor and associated drive transmission equipment. Resistance losses in the motor, which rise with temperature, would increase. Providing adequate ventilation and keeping motor cooling ducts clean can help dissipate heat to reduce excessive losses. The life of the insulation in the motor would also be longer: for every 10°C increase in motor operating temperature over the recommended peak, the time before rewinding would be needed is estimated to be halved

A checklist of good maintenance practices to help insure proper motor operation would include:

- Inspecting motors regularly for wear in bearings and housings (to reduce frictional losses) and for dirt/dust in motor ventilating ducts (to ensure proper heat dissipation).
- Checking load conditions to ensure that the motor is not over or under loaded. A change in motor load from the last test indicates a change in the driven load, the cause of which should be understood.
- Lubricating appropriately. Manufacturers generally give recommendations for how and when to lubricate their motors. Inadequate lubrication can cause problems, as noted above. Over lubrication can also create problems,
- e.g. excess oil or grease from the motor bearings can enter the motor and saturate the motor insulation, causing premature failure or creating a fire risk.
- Checking periodically for proper alignment of the motor and the driven equipment. Improper alignment can cause shafts and bearings to wear quickly, resulting in damage to both the motor and the driven equipment.
- Ensuring that supply wiring and terminal box are properly sized and installed. Inspect regularly the connections at the motor and starter to be sure that they are clean and tight.

Age

Most motor cores in India are manufactured from silicon steel or de-carbonized cold-rolled steel, the electrical properties of which do not change measurably with age. However, poor maintenance (inadequate lubrication of bearings, insufficient cleaning of air-cooling passages, etc.) can cause a deterioration in motor efficiency over time. Ambient conditions can also have a detrimental effect on motor performance. For example, excessively high temperatures, high dust loading, corrosive atmosphere, and humidity can impair insulation properties; mechanical stresses due to load cycling can lead to misalignment. However, with adequate care, motor performance can be maintained.

Example2.1:

A three phase,10 kW motor has the name plate details as 415 V, 18.2 amps and 0.9 PF. Actual input measurement shows 415 V, 12 amps and 0.7 PF which was measured with power analyzer during motor running. Determine the motor loading?

Rated output at full load = 10 kW

Rated input at full load = □**3 x V x I x CosΦ** = 1.732 x 0.415 x 18.2 x 0.9 = 11.8 kW

Example 2.2:

A 400 Watt mercury vapor lamp was switched on for 10 hours per day. The supply volt is 230 V. Find the power consumption per day? (Volt = 230 V, Current = 2 amps, PF = 0.8)

Electricity consumption (kWh) = V x I x CosΦ x No of Hours

= 0.230 x 2 x 0.8 x 10 = 3.7 kWh or Units

Example 2.3:

An electric heater of 230 V, 5 kW rating is used for hot water generation in an industry. Find electricity consumption per hour (a) at the rated voltage (b) at 200 V.

(a) Electricity consumption (kWh) at rated voltage = 5 kW x 1 hour = 5 kWh.

(b) Electricity consumption at 200 V (kWh) = (200 / 230)2 x 5 kW x 1 hour = 3.78 kWh.

Example2.4 :

The utility bill shows an average power factor of 0.72 with an average KW of 627. How much kVAr is required to improve the power factor to .95 ?

Using formula

Cos Φ 1 = 0.72 , tan Φ 1 = 0.963

Cos Φ 2 = 0.95 , tan Φ 2 = 0.329

kVAr required = P (tan Φ 1 - tan Φ 2)

= 627 (0.964 – 0.329)

= 398 kVAr

7. **What is the Need for Electrical Load Matching? (5)**

In a macro perspective, the growth in the electricity use and diversity of end use segments in time of use has led to shortfalls in capacity to meet demand. As capacity addition is costly and only a long-time prospect, better load management at user end helps to minimize peak demands on the utility infrastructure as well as better utilization of power plant capacities. The utilities (State Electricity Boards) use power tariff structure to influence end user in better load management through measures like time of use tariffs, penalties on exceeding allowed maximum demand, night tariff concessions etc. Load management is a powerful means of efficiency improvement both for end user as well as utility. As the demand charges constitute a considerable portion of the electricity bill, from user angle too there is a need for integrated load management to effectively control the maximum demand.

8. **Enumerate the peak demand control methodologies used in energy management planning (6)**

Step by Step Approach for Maximum Demand Control

Load Curve Generation

Presenting the load demand of a consumer against time of the day is known as a 'load curve'. If it is plotted for the 24 hours of a single day, it is known as an 'hourly load curve' and if daily demands plotted over a month, it is called daily load curves. A typical hourly load curve for an engineering industry is shown in Figure 1.5. These types of curves are useful in predicting patterns of drawl, peaks and valleys and energy use trend in a section or in an industry or in a distribution network as the case may be.

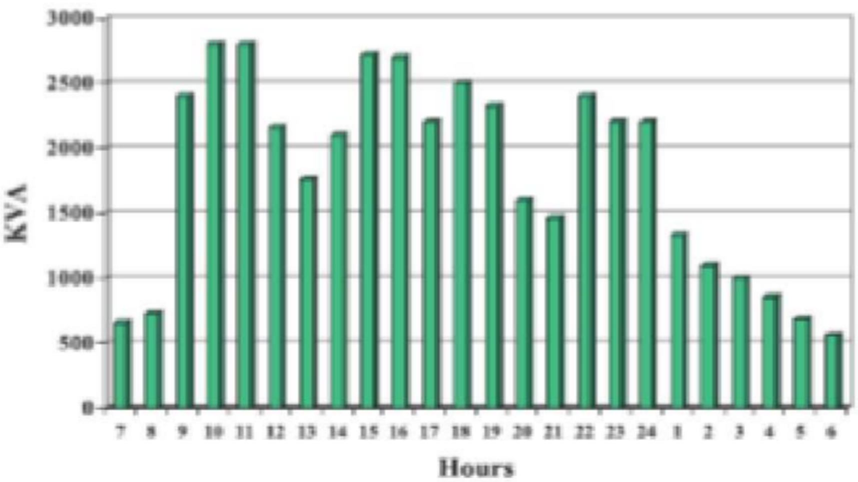


Figure 1.5 Maximum Demand (Daily Load Curve, Hourly kVA)

Rescheduling of Loads

Rescheduling of large electric loads and equipment operations, in different shifts can be planned and implemented to minimize the simultaneous maximum demand. For this purpose, it is advisable to prepare an operation flow chart and a process chart. Analyzing these charts and with an integrated approach, it would be possible to reschedule the operations and running equipment in such a way as to improve the load factor which in turn reduces the maximum demand.

Storage of Products/in process material/ process utilities like refrigeration

It is possible to reduce the maximum demand by building up storage capacity of products/ materials, water, chilled water / hot water, using electricity during off peak periods. Off peak hour operations also help to save energy due to favorable conditions such as lower ambient temperature etc. Example: Ice bank system is used in milk & dairy industry. Ice is made in lean period and used in peak load period and thus maximum demand is reduced.

Shedding of Non-Essential Loads

When the maximum demand tends to reach preset limit, shedding some of non-essential loads temporarily can help to reduce it. It is possible to install direct demand monitoring systems, which will switch off non-essential loads when a preset demand is reached. Simple systems give an alarm, and the loads are shed manually. Sophisticated microprocessor controlled systems are also available, which provide a wide variety of control options like:

- Accurate prediction of demand
- Graphical display of present load, available load, demand limit
- Visual and audible alarm
- Automatic load shedding in a predetermined sequence
- Automatic restoration of load
- Recording and metering
- Operation of Captive Generation and Diesel Generation Sets

When diesel generation sets are used to supplement the power supplied by the electric utilities, it is advisable to connect the D.G. sets for durations when demand reaches the peak value. This would reduce the load demand to a considerable extent and minimize the demand charges.

Reactive Power Compensation

The maximum demand can also be reduced at the plant level by using capacitor banks and maintaining the optimum power factor. Capacitor banks are available with microprocessor-based control systems. These systems switch on and off the capacitor banks to maintain the desired Power factor of system and optimize maximum demand thereby.

9. Explain selection of motors for constant and variable loads. (7)

Types of Motor Loads

The type of load that a motor drives is one of the most important application considerations when applying any type of AC drive.

For some types of loads, the application considerations may be minimal. For other types of loads, an extensive review may be required.

Generally, loads can be grouped into three different categories:

- Constant Torque Loads – conveyors, hoists, drill presses, extruders, positive displacement pumps (torque of these pumps may be reduced at low speeds).
- Variable Torque Loads – fans, blower, propellers, centrifugal pumps.
- Constant Horsepower Loads – grinders, turret lathes, coil winders.

Constant Torque Loads

Constant torque loads are where applications call for the same amount of driving torque throughout the entire operating speed range.In other words, as the speed changes, the load torque remains the same.

Constant torque applications include everything that is not variable torque applications. In fact, almost everything but centrifugal fans and pumps are constant torque.

Variable Torque Loads

As was just mentioned, there are only two kinds of variable torque loads: centrifugal pumps and fans.

With a variable torque load, the loading is a function of the speed. Variable torque loads generally require low torque at low speeds and higher torque at higher speeds.

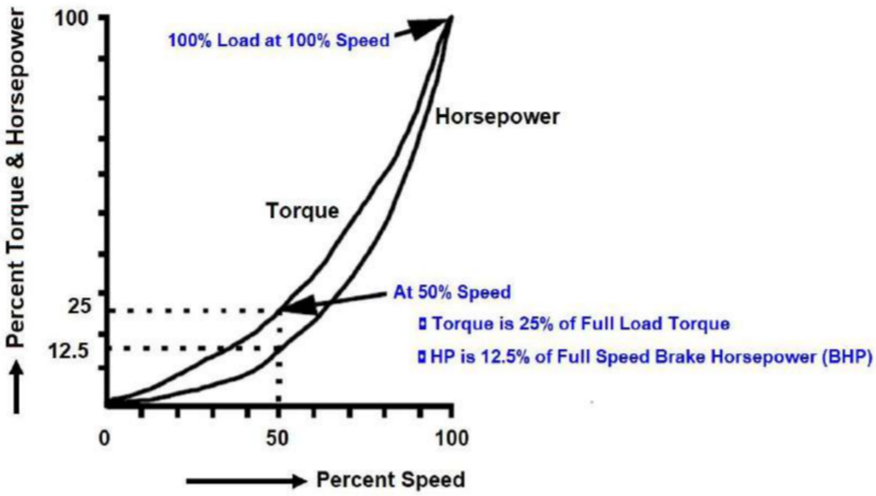
Fans and pumps are designed to make air or water flow. As the rate of flow increases, the water or air has a greater change in speed put into it by the fan or pump, increasing its inertia.

In addition to the inertia change, increased flow means increased friction from the pipes or ducts. An increase in friction requires more force (or torque) to make the air or water flow at that rate.

The effects that reduced speed control has on a variable torque fan or pump are summarized by a set of rules known as the Affinity Laws.

The basic interpretation of these laws is quite simple:

- The flow produced by the device is proportional to the motor speed.
- The pressure produced by the device is proportional to the motor speed squared.
- The horsepower required by the device is proportional to the motor speed cubed.



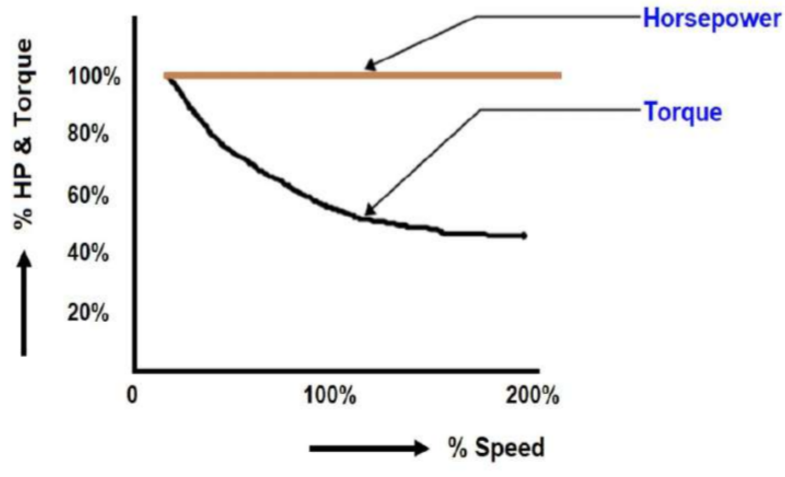
The cube law (third item) load is at the heart of energy savings. The change in speed is equal to the horsepower cubed.

For example, you might expect a 50% change in speed would produce a 50% change in volume and would require 50% of the horsepower.

Luckily for us, this 50% change in speed must be cubed, representing only 12.5% of the horsepower required to run it at 100% speed.

The reduction of horsepower means that it costs less to run the motor. When these savings are applied over yearly hours of operation, significant savings accumulate.

Constant Horsepower Loads



A constant horsepower load is when the motor torque required is above the motor's base speed (60 Hz). With a constant horsepower type of load, the torque loading is a function of the changing physical dimensions of the load.

These types of applications would include grinders, lathes, and winding reels. Constant horsepower loads require high torque at low speeds and low torque at high speeds.

While the torque and speed changes, the horsepower remains constant.

For example, an empty reel winding a coil will require the least amount of torque, initially, and will be accelerated to the highest speed. As the coil builds up on the reel, the torque required will increase and the speed will be decreased.

10. Explain Present maximum efficiency standards for power and distribution transformers. (10)

Indian Standard on Distribution Transformers was first issued in 1958, covering distribution transformers up to 100 kVA. The first part of the fourth revision to this standard came out in 2014 with 4 amendments till 2021, covering transformers up to 2500 kVA 33 kV. This is titled "Outdoor/Indoor Type Liquid Immersed Distribution Transformers up to and including 2500 kVA 33kV -Specification- Part 1: Mineral Oil Immersed. Part 3 of the standard, "Natural/Synthetic Organic Ester liquid Immersed "was published in 2021.

1. **Reducing the variety of Power (kVA) Ratings.**

The rated power of distribution transformers is selected from the R10 number series given in ISO 3:1973 Preferred Numbers-Series of preferred numbers. It means ratings between 1-10 are in ten steps each incremental rating being multiplied by 1.26 (the tenth root of 10) ie 1-1.25 -1.6 -2 – 2.5 -3.15- 4-5 – 6.3- 8- 10 kVA or MVA and their multiples by 10 &100. This results in a large variety of transformer sizes. If we use the R5 series for selection (5 steps in multiples of the fifth root of 10=1.58),the sizes will be 1-1.6 – 2.5 – 4 – 6.3- 10 kVA or MVA, and multiples of 10 &100 for higher ratings. This way the kVA ratings of distribution transformers can be halved.

2. **Permissible Working Flux Density**

In the IS 1180-1, under clauses 6.9 & 7.9 the following stipulation is given:
Clause 7.9.1 The maximum flux density in any part of the core and yoke at rated voltage and frequency shall be such that the flux density with + 12.5 percent combined voltage and frequency variation from rated voltage and frequency does not exceed 1.9 Tesla.

3. **Energy Efficiency Levels.**

IEC standard 60076-20:2017" Power Transformers Energy Efficiency" proposes three methods for evaluating the energy performance of a transformer. These are based on the practices followed in different countries of the world.

- the Peak Efficiency Index (PEI) should be used in conjunction with either a total cost of ownership (TCO) approach or any other means of specifying the load factor.
- the no-load and load losses at rated power for rationalization of transformer cores and coils for transformers generally produced in large volumes;
- the efficiency at a defined power factor and particular load factor (typically at 50 %).

4. **Separate Standard for Ester filled Transformers**

Part 3 of IS 1180 was issued in 2021, titled "Outdoor/ Indoor Type Liquid Immersed distribution transformers up to and including 2500 kVA 33 kV -Specifications -Part 3 Natural/ Synthetic Organic Ester liquid Immersed "Transformer performance parameters are not affected by the insulating fluid used, whether esters or silicon oil and IEC and other National Standards on Power Transformers do not have a separate standard for ester-filled transformers. The standards are titled Liquid immersed transformers to cover any type of insulating fluids. Hence this Part 3 is redundant and may be withdrawn. Part 1 may be revised from Mineral Oil-immersed to Liquid Immersed Transformers.

5. **Auto- transformers up to and including 420 kV**:

At present in the country there are 220 kV class auto-transformers of different tapping arrangement

a) Taps on HV (series winding) for HV variation by +- 10% with two options- constant ohmic or constant percentage impedance.

b) Taps on MV line end for MV variation by +-10 % with two options- constant ohmic or constant percentage impedance.

As standard, HV (series winding) taps for HV variation with constant percentage impedance is selected.

With respect to 400 kV class auto-transformers, there are two voltage ratios- 400/ 220 kV (most common) and 400/132 kV. In case of 400/132 kV class transformers, OLTC (On Load tap Changer) is provided in HV (series) winding for HV variation by +-10 % with regulating winding positioned between tertiary and common so as to get constant ohmic impedance at all taps. But an economical solution used in most countries is to provide OLTC at neutral end of common winding for MV variation. **This neutral end application of OLTC is selected as the standard.** Hence **as standard, fixed ratio 400/220 kV auto-transformers without OLTC is selected as standard and where ever OLTC is essential, constant percentage impedance arrangement is proposed as standard.**

6. **800 kV Auto-Transformers**

When the first 3x500 MVA 765/400 kV auto-transformers were procured in 2005, these were double limb wound with OLTC on HV neutral end for HV variation by +-5 %. But transformers procured in later years were single limb wound type with tertiary and regulating windings on the side core limb. These were with very high tertiary impedances (HV-Try 195 % and MV-Try with 180 % instead of the earlier 67 % and 49 % respectively) and at present these high tertiary impedance transformers are the standard. Units from these two types of banks cannot be interchanged to form three phase banks, due to the difference in tertiary impedance. **In future, at many locations fixed ratio units will be sufficient and where ever, OLTC is required, single limb Try-Reg-Common-Series winding arrangement is proposed.**

Purchaser has the right to check the specific loadings (current density in winding and flux density in core) at various stages of procurement (tender stage, design review stage, manufacturing stage) against the values furnished in GTP

Stabilizing delta tertiary winding is not required for transformers with 3 phase 3 limbed core construction. At present, stabilizing delta tertiary winding is not provided in YNyn connected transformers up to a rating of 100 MVA. Since rating has no significance with stabilizing tertiary, core construction is considered as criteria for deciding the necessity of stabilizing delta winding.

11.Explain design measures for increasing efficiency in electrical system components. (7)

Maximum Demand Controllers

High-tension (HT) consumers have to pay a maximum demand charge in addition to the usual charge for the number of units consumed. This charge is usually based on the highest amount of power used during some period (say 30 minutes) during the metering month. The maximum demand charge often represents a large proportion of the total bill and may be based on only one isolated 30 minute episode of high power use.

Automatic Power Factor Controllers

Various types of automatic power factor controls are available with relay / microprocessor logic. Two of the most common controls are: Voltage Control and kVAr Control.

Voltage Control

Voltage alone can be used as a source of intelligence when the switched capacitors are applied at point where the circuit voltage decreases as circuit load increases. Generally, where they are applied the voltage should decrease as circuit load increases and the drop in voltage should be around 4 – 5 % with increasing load.

KILOVAR Control

Kilovar sensitive controls (see Figure 10.2) are used at locations where the voltage level is closely regulated and not available as a control variable. The capacitors can be switched to respond to a decreasing power factor as a result of change in system loading.

Automatic Power Factor Control Relay

It controls the power factor of the installation by giving signals to switch on or off power factor correction capacitors. Relay is the brain of control circuit and needs contactors of appropriate rating for switching on/off the capacitors.

Intelligent Power Factor Controller (IPFC)

This controller determines the rating of capacitance connected in each step during the first hour of its operation and stores them in memory. Based on this measurement, the IPFC switches on the most appropriate steps, thus eliminating the hunting problems normally associated with capacitor switching.

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